

NITIN JHA

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[GitHub](#) | [LinkedIn](#) | [HackerRank](#) | [LeetCode](#)

EDUCATION

ABES Engineering College

B.Tech in Electrical and Computer Engineering

Ghaziabad, India

Nov 2022 – Present

Kendriya Vidyalaya

Senior Secondary (PCM with Computer Science)

Percentage: 83.34%

Delhi, India

July 2022

TECHNICAL SKILLS

- **Industrial Automation:** Modbus RTU Protocol, VFD Hardware Interfacing, Awareness of IEC 61850, Telemetry Logging.
- **Machine Learning:** Random Forest, XGBoost, Feature Engineering, Standard Scaling, Stratified Sampling, Hyperparameter Tuning.
- **Software Development:** Python (Pandas, Scikit-learn, joblib), C, SQL, CustomTkinter (GUI Builder), Regex for Log Parsing.
- **Expert Systems:** Integration with Large Language Models (Google Gemini AI) for automated diagnostic reporting.

PROJECTS / OPEN-SOURCE

Predictive Maintenance for Electrical Machines

Python, Modbus, Scikit-learn, XGBoost, CustomTkinter

Industrial Telemetry & ML-Driven Diagnostics

- Engineered a real-time data acquisition pipeline using **Modbus RTU** (via `minimalmodbus`) to monitor induction motors controlled by **Variable Frequency Drives (VFD)**.
- Developed advanced physical features for fault detection, including **Cooling Efficiency ($^{\circ}\text{C}/\text{VA}$)**, **Load Friction Index (A/Hz)**, and **Apparent Power (S)**.
- Analyzed and classified 4 machine states with **$\sim 88\%$ accuracy**: Healthy (Nominal), Overheat ($\text{Temp} > 95^{\circ}\text{C}$), Unbalanced Load (15% current rise), and Bearing Faults (erratic spikes $> 4.0\text{A}$).
- Integrated **Google Gemini AI (LLM)** to provide real-time, context-aware maintenance consultation and corrective measures based on sensor anomalies.

BookBuddy: Personalized Recommendation Chatbot

Python

ML-Based Natural Language Consultation

- Built a recommendation engine solving the challenge of personalized book discovery using user history.
- Developed an ML-integrated chatbot that suggests books with persistent chat history and reviews.
- Improved user engagement through personalized result filtering and interactive conversational picking.

IntViu: AI-Powered Mock Interview Platform

Next.js, Tailwind CSS, Vapi SDK, Shadcn UI

Real-Time Voice-Interactive Platform

- Developed a voice-interactive AI platform using Next.js and Vapi SDK to automate mock interview cycles.
- Implemented real-time voice feedback systems, enabling 24/7 personalized practice for technical candidates.
- Leveraged Shadcn UI and Tailwind CSS for a high-fidelity, responsive user interface.

WORK EXPERIENCE

Fiverr | Freelance Python Developer

Remote | 2023 – 2024

Automated data analysis pipelines and telemetry dashboards for international clients using Python (Pandas, Matplotlib) and SQL. Specialized in converting raw sensor logs into visual management reports.

CERTIFICATIONS & HONORS

- **Data Analytics With Python** – NPTEL (Swayam)
- **Google Cloud Computing Foundations** – NPTEL
- **Top 100th Position** – GDSC Solution Challenge 2024
- **Data Science Virtual Experience Programme** – British Airways